



11. To find the full load quantities, draw line BK (= full load output / power scale). Now, draw line PK parallel to output line meeting the circle at point P.

12. Draw line PT parallel to Y-axis meeting output line at Q, torque line at R, constant loss line at S and X-axis at T.

Note: Choose the current scale such that the circle diagram will be as large as possible. The larger the circle diagram more will be the accuracy. Select power scale = $\sqrt{3} \times V_{rated} \times \text{current scale}$.

TABULATION

NO LOAD TEST					BLOCKED ROTOR TEST				
V _{oc}	I _{oc}	W ₁	W ₂	W _{oc}	V _{sc}	I _{sc}	W ₁	W ₂	W _{sc}

Stator resistance measurment			
S.No.	V (volts)	I (amps)	R _{dc} =V/I Ω
1.			
2.			
3.			
4.			
R _{dc} =			

CIRCLE DIAGRAM

V_{oc} = 400V. I_{oc} =A, W_{oc} =W

V_{sc} =V. I_{sc} =V. W_{sc} =W

Per phase values are

V_o = V_{oc} =V

$$I_0 = \frac{I_{oc}}{\sqrt{3}} =A$$

V_s = V_{sc} =V

$$I_s = \frac{I_{sc}}{\sqrt{3}} =A$$